

Chapter 4- Linear Equations

4.1 Introduction

An equation is a statement that two expressions are equal.

A linear equation in one variable has the general form:

$$ax + b = 0 \quad \text{where } a \text{ is not equal to } 0$$

The value of x that satisfies the equation is called its solution or root.

Example 1: Solve $3x + 5 = 20$

$$3x = 20 - 5$$

$$3x = 15$$

$$x = 5$$

4.2 Solving Equations

To solve a linear equation we perform the same operation on both sides until the variable stands alone.

Rules:

1. Add or subtract the same number on both sides.
2. Multiply or divide both sides by the same non-zero number.

Example 2: Solve $2(x - 3) = 10$

$$2x - 6 = 10$$

$$2x = 16$$

$$x = 8$$

Check: $2(8 - 3) = 2(5) = 10$ Correct.

4.3 Word Problems

Many real problems become linear equations once we name the unknown quantity.

Example 3:

The sum of two consecutive numbers is 47. Find them.

Let the first number be x .

Then the next number is $x + 1$.

$$x + (x + 1) = 47$$

$$2x + 1 = 47$$

$$2x = 46$$

$$x = 23$$

The numbers are 23 and 24.

4.4 Practice Exercises

Solve for x:

1. $x + 7 = 15$
2. $4x = 36$
3. $2x - 9 = 11$
4. $5(x + 2) = 40$
5. $3x + 4 = x + 18$

Word problems:

6. A number multiplied by 4 gives 52. Find it.
7. The perimeter of a square is 64 cm.
Find the length of one side.

4.5 Summary

Key points from this chapter:

- A linear equation has the form $ax + b = 0$.
- The highest power of the variable is 1.
- Whatever is done to one side must be done to the other.
- Always verify the answer by substitution.

Next chapter: Linear Equations in Two Variables.